

Planning and action selection

Project: Grounded Learning Service

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

State machines for bounded agents

Without opening the old lesson, explain one key idea from State machines for bounded agents. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Failure budgets and service objectives

Without opening the old lesson, explain one key idea from Failure budgets and service objectives. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Testing model-dependent behaviour

Without opening the old lesson, explain one key idea from Testing model-dependent behaviour. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Planning and action selection. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Planning and action selection. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Plan validates input then sends a message before checking recipient approval. The task authorises drafting only. Locate violation.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

Generated plan retrieves a room, books it and then answers. It calls booking necessary for helpfulness. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: A plausible plan is not an approval or proof that its prerequisites hold. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

Plan validates input then sends a message before checking recipient approval. The task authorises drafting only. Locate violation.

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled planning and action selection evidence sheet to Grounded Learning Service. Show how you can explain the mechanism and terminology of planning and action selection. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Planning and action selection

Plan around missing evidence. A bot must answer a tower-height question using approved cards only.

Search finds no height card. Allowed actions: search, clarify, return unsupported. Unapproved web access is excluded.

Planning and action selection

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Name missing input

A supported height answer needs an approved height record.



2. Select allowed action

After the empty search, return unsupported rather than inventing a number.



3. Explain next step

Ask for an approved source card if the question should be answered.

CHANGE ONE THING

Supply an approved card giving height 24 metres.

Explain the changed result using the matching lesson.

Choose a permitted plan when no approved tower-height card exists. What new answer becomes supported after a card records 24 metres, and why could the bot not invent it earlier?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

13. Project portfolio milestone

Project: Grounded Learning Service

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Plan around missing evidence. A bot must answer a tower-height question using approved cards only.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Ask for an approved source card if the question should be answered.

1. Select allowed action
2. Explain next step
3. Name missing input

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Explain next step Ask for an approved source card if the question should be answered. Planning selects useful actions within available evidence and authority. More steps cannot create permission to access an excluded source.

Practice 2: explain the deciding evidence

Changed-case practice: Supply an approved card giving height 24 metres. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. The bot can retrieve it and answer 24 metres with attribution.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The bot can retrieve it and answer 24 metres with attribution. The missing factual input is now available within the allowed collection.

Repair a missing building block

Review State machines for bounded agents (ai-6/13). Goal: Explain the mechanism and terminology of state machines for bounded agents. Short practice: Trace A and B through the state diagram. Name B’s forbidden transition and explain whether adding checking→querying permits that shortcut.. Return to this week and explain what you changed.

Review Failure budgets and service objectives (ai-6/11). Goal: Explain the mechanism and terminology of failure budgets and service objectives. Short practice: Calculate the maximum allowed failures among 100 requests. Does 97 successes and 3 failures meet the target, and what changes after one failure becomes a success?. Return to this week and explain what you changed.

Review Testing model-dependent behaviour (ai-6/8). Goal: Explain the mechanism and terminology of testing model-dependent behaviour. Short practice: Score "Friday in room B" and "Friday in room D" against both mandatory criteria. Would a correct 25-word response pass, and why?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
